

Research paper

Aerothermodynamic study of Two-Stage-To-Orbit system composed of wide-speed-range vehicle and rocket

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ABSTRACT

Hypersonic vehicles with Mach numbers greater than five exhibit complex shock wave interactions, accompanied by locally enhanced heat flux near the shock-affected surface. To investigate the unclear interstage thermal environment in a Two-Stage-To-Orbit system, a model composed of a wide-speed-range vehicle and a reusable rocket is employed in an aerothermodynamic study of the Two-Stage-To-Orbit system considering the interstage interactions performed at freestream Mach number = 6. The innovations of this paper are the complex model, which is highly similar to the real situation, and the study of the thermal environment between stages near different reflection positions, whereby the shock wave interference that causes different forms of heat flux is divided into two categories. The thermal design of the Two-Stage-To-Orbit system should especially consider the head of the wide-speed-range vehicle, interstage distance, and the swept angle of canted fins and wings, as well as their width. This paper adopts the second-order AUSMDV (a variant of the Advection Upstream Splitting Method) scheme for spatial discretization and Lower–Upper Symmetric Gauss–Seidel method for time discretization in formulating the Reynolds-averaged Navier–Stokes equations, which are then solved by the finite volume method and Menter's shear stress transport $k-\omega$ turbulence model. Numerical analysis sheds light on the complex shock structure and the reasons for the considerable increase in wall temperature and aerothermal loads at critical parts of various reflected shock positions. The incident shock wave generated by the head apex of the rocket constantly reflects and gradually weakens between the two stages, which brings about shock wave/boundary-layer interaction and gives rise to elevated heat flux. The moderate incident shock and the weak reflected shock are primarily responsible for the lack of boundary-layer separation, although the expansion caused by the convex curvature of the head of the wide-speed-range vehicle also contributes. A non-reflected part of the incident shock wave interacts with the windward surface of the canted fins and wings, producing shock wave/boundary-layer interaction and a V-shaped heat flux distribution that extends to the leading edges of the wings. As the angle of attack decreases, the intensity of the incident shock and the reflected shock gradually weakens, leading to reduced heat flux on the windward surface of the wide-speed-range vehicle. The range over which the incident shock impinges the walls also narrows, corresponding to the smaller heat flux distribution. These aerothermal studies comprehensively determine the law of thermal changes, which is significant for the design of thermal protection in Two-Stage-To-Orbit systems.

1. Introduction

Hypersonic vehicles with Mach numbers greater than five are becoming a reliable and feasible way of achieving long-distance and fast flight and global space travel. During climbing and re-entry, the air viscosity generates high levels of aerodynamic heat on the hypersonic vehicle, which can seriously affect its safety [1,2]. Hence, a single-stage-to-orbit system must simultaneously enable in-atmosphere maneuvering, in-orbit propulsion, in-orbit transfer, and atmospheric re-entry, which present significant difficulties for the design of the

center of gravity and the balance between fuel and other equipment. These difficulties are simplified in Two-Stage-To-Orbit (TSTO) systems, in which the flight path is decomposed into two stages. The first stage adopts a reusable rocket [3–7] and a space shuttle to boost the hypersonic vehicle. The heavy booster is then discarded from the light vehicle body and safely returns to the ground for secondary use. The idea of TSTO helps to reduce the long-term cost of the launch cycle, but introduces additional aerodynamic heating problems between the booster and the vehicle. These problems, which are caused by the

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Nomenclature

TSTO	Two-Stage-To-Orbit
TPS	Thermal protection system
NS	Navier–Stokes
RANS	Reynolds Averaged Navier–Stokes
LU-SGS	Lower–Upper Symmetric Gauss–Seidel
AUSMDV	A variant of the Advection Upstream Splitting Method
WSRV	Wide-speed-range vehicle
SWBLI	Shock wave/boundary-layer interaction
LGBB	Langley Glide-Back Booster
FP	Flat plate
HCC	Hemispherical circular cylinder
CFD	Computational fluid dynamics
SST	Shear stress transport
y+	Yplus above viscous walls
q_{ref}	Heat flux reference value
q/q_{ref}	Heat flux ratio, referred to as heat flux
Ma_{∞}	Freestream Mach number
Re_{∞}	Freestream unit Reynolds number
P_{∞}	Freestream pressure
T_{∞}	Freestream temperature
T_w	Wall temperature
C_p	Pressure coefficient
C_f	Skin friction coefficient
CFL	Courant–Friedrichs–Levy number
$\nabla \rho$	Density gradient
W	Conserved variables
F_c	Inviscid fluxes
F_v	Viscous fluxes
ρ	Density
u, v, w	Velocity in the x, y, z
V_n	Normal velocity of the control body surface
n_x, n_y, n_z	Unit normal vector of the control body surface
p	Pressure
H	Total enthalpy per unit mass
E	Total energy per unit mass
Θ	Heat conduction in the fluid
k	Thermal conductivity coefficient
q_w	The vector of wall heat flux
μ	Molecular viscosity
γ	Specific heat ratio
Pr	Prandtl number
∞	Freestream
w	wall
∇	Hamiltonian operator

reflection and interaction of shock waves, require further investigation to provide peak thermal loads for the design of the thermal protection system (TPS) [4].

To date, several studies have focused on the aerodynamics of TSTO, but there has been little research on the aerothermodynamics of TSTO systems composed of simplified models. In terms of aerodynamic research, Bordelon et al. [8] conducted a wind tunnel test of the separation between two stages of a conventional TSTO launch vehicle consisting of a Langley Glide-Back Booster (LGBB) system, thus determining the interfering aerodynamic forces and the short-range flow environment. Liever [9] further verified the simulation method in

dynamic simulations of the separation between the two stages of a TSTO launch vehicle incorporating LGBB. Derek [10] investigated the aero-heating characteristics of the same configuration of LGBB in a Mach 6 air tunnel, and suggested that the heating of the upwind centerline increases rapidly when the orbiter is separated from the booster. An aerodynamic study of a TSTO system considering various boosters in hypersonic flow was performed by Uematsu et al. [11]. The rocket was simplified into a triangular flat plate (FP) and the orbiter into a hemispherical circular cylinder (HCC), which cannot adequately simulate the real thermal environment. An experimental study of the unsteady aerodynamic interaction between an FP and an HCC at $M_{\infty} = 8.1$ was studied by Ozawa et al. [12,13]. The results show that the clearance between two bodies has a significant effect on the flow field of the aerodynamic heating of TSTO systems. Kitamura et al. [14] analyzed a TSTO model composed of a delta wing and a hemispherical cylinder in terms of the aerodynamic interference and the mechanism of high heat flux at the nose. The results suggest that the heat flux increment varies with distance. The aerodynamic interference of a new type of conceptual TSTO model consisting of a trapezoid wing and a hemisphere-cone-cylinder wing, based on the large-eddy simulation/Reynolds Averaged Navier–Stokes (RANS) method, was investigated by Jia et al. [15], with the results showing that new features are produced by the wings added to the orbiter. In short, most research has focused on the aerodynamic problems of the TSTO, and few studies have examined the heat flux at the nose or the influence of the two-stage distance on the heat flux using simple models. However, it is necessary to study the aerothermodynamic behavior of models that are similar to real configurations, because the shape of the aircraft has a strong influence on the aerodynamic heating of the orbiter.

In terms of aerothermodynamics, researchers have mainly considered a single vehicle in experimental research and numerical simulations. On the experimental side, Hollis et al. [16] reported a hypersonic heating study of the X-33 TPS at NASA Langley Research Center. Experimental data from the Mars Pathfinder entry were used to analyze the environment and material response by Milos et al. [17], and a combination of calculations, predictions, and experimental measurements of the expected aerodynamic heating in the process of two microprobes entering Mars was discussed by Mitcheltree et al. [18]. As there are some issues with experimental research, such as the experimental conditions and high experimental cost, computational fluid dynamics (CFD) is often used as an economical and effective alternative. Thus, there has been considerable aerothermodynamic research on spacecraft using CFD. Numerical simulations of the aerothermal characteristics of the swept leading edge of a hypersonic vehicle were performed by Kumar et al. [19], leading to the proposal of aerothermal concepts such as temperature and drag-minimized sweepback and the thermally benign sharp swept-back leading edge effect. Knight et al. [20] reported a series of independent computations regarding two configurations (double cone and cylinder), while Hamilton et al. [21] computed heating rates for various body shapes, including a spherically blunted cone, the X-33, a capsule configuration similar to the crew exploration vehicle, and a shuttle orbiter. Sohail et al. [22] presented CFD results for various models, such as a half-cylinder, cone flare, hypersonic cone, and cone cylinder frustum, with aero-heating at high Mach numbers, and predictions of the aerothermodynamic behavior of a conceptual aerospace configuration were presented by Kinney et al. [23]. Numerical simulations of aerodynamic characteristics around a small hypersonic plane under various flow conditions were carried out by Oriano et al. [24]. They investigated the effects of the elevon deflection and the presence of the engines on aerodynamic performance, and analyzed the effects of the boundary-layer laminar–turbulent transition on the aero-heating behavior. Convection coefficients and heat fluxes on hypersonic vehicles were analyzed by Mahulikar et al. [25], and the applicability of the recovery temperature in the stagnation zone was discussed. Computations of a blunted cone flare for laminar flow at Mach 6 were presented by Savino et al. [26], while an uncertainty

analysis of laminar and turbulent aero-heating predictions for a 70° spherically blunted cone at the peak heating condition during Mars entry was performed by Wang et al. [27]. The influence of geometric heatshield parameters on the aerothermodynamic performance of a Mars entry vehicle was studied by Ju et al. [28]. The TPS on a blunt body combining spike technology with opposing jet technology has been investigated by Qu [29] and Ou [30], and the numerical results were found to be in good agreement with the measured pressure and surface heat flux distribution in an H3 wind tunnel. The hypersonic environment characteristics of the CIRA FTB-X re-entry vehicle and a trajectory-based design approach were investigated by Pezzella [31], who used several engineering and CFD analyses. Despite the extensive investigations mentioned above, there has been no research on the thermal problem of the interstage interference of the TSTO. Thus, this issue requires further study.

Most researchers have focused on predicting the thermal environment of a single vehicle and a simplified TSTO model, and the complexity of the real model has rarely been considered. The aerothermodynamic performance of the real TSTO system contains more serious thermal environment problems than those of a single vehicle [5] and the simplified TSTO system, including complex shock wave interference, and which gives rise to high thermal loads due to the distance between the aircraft is very close. As the thermal environment and flow field of real parallel vehicles are not, at present, sufficiently clear, this paper reports on CFD computations for a TSTO system composed of a wide-speed-range vehicle (WSRV) and a rocket with wings. The objective of this study is to understand the intense thermodynamic performance for a more complex configuration with wings, and to analyze some of the critical parts of various reflected shock positions with local amplification of heating. Hence, we mainly focus on the aerothermal phenomena at three reflected shock positions involving the windward surfaces of the WSRV and rocket, and the leading edges of the wing around the reflected shock. Finally, the influence of the angle of attack on the interstage thermal environment is also studied.

2. Numerical simulation

2.1. Geometry definition and grid generation

The WSRV–rocket geometry is shown in Fig. 1, and consists of a WSRV fuselage, wing, vertical tail, and rocket with wings. The geometry of the WSRV refers to a quarter-scale SR72 [32–35]. The length, width, and height of the WSRV are 9.5 m, 5.2 m, and 0.9 m, respectively, and those of the rocket are 10.5 m, 6.2 m, and 1 m. The staging distance between the WSRV and the rocket is 0.38 m. An unstructured grid is used to calculate the TSTO model, with a structured grid used near the wall for normalization. Thus, y^+ in the first layer of the grid is less than 1, and the number of grid elements is 63.3 million.

2.2. Governing equations

The compressible NS equations in the form of a three-dimensional integration [36] are used for the aerothermal numerical simulations:

$$\frac{\partial}{\partial t} \int_{\Omega} \mathbf{W} d\Omega + \oint_{\partial\Omega} (\mathbf{F}_c - \mathbf{F}_v) dS = 0 \quad (1)$$

where $\mathbf{W}$, $\mathbf{F}_c$, and $\mathbf{F}_v$ are the conserved variables, inviscid fluxes, and the viscous fluxes, respectively. These are expressed as follows:

$$\mathbf{W} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ \rho E \end{pmatrix} \quad (2)$$

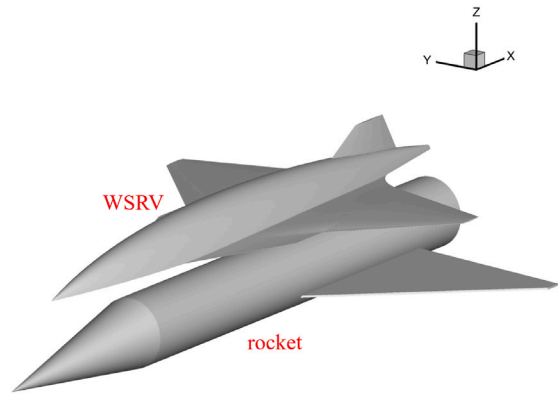


Fig. 1. Schematics of WSRV–rocket system.

$$\mathbf{F}_c = \begin{pmatrix} \rho V_n \\ \rho u V_n + n_x p \\ \rho v V_n + n_y p \\ \rho w V_n + n_z p \\ \rho H V_n \end{pmatrix} \quad \mathbf{F}_v = \begin{pmatrix} 0 \\ n_x \tau_{xx} + n_y \tau_{xy} + n_z \tau_{xz} \\ n_x \tau_{yx} + n_y \tau_{yy} + n_z \tau_{yz} \\ n_x \tau_{zx} + n_y \tau_{zy} + n_z \tau_{zz} \\ n_x \Theta_x + n_y \Theta_y + n_z \Theta_z \end{pmatrix} \quad (3)$$

where ρ denotes the density, u , v , w represent the velocity in the x , y , z directions, respectively, p represents the pressure, E is the total energy per unit mass, and H is the total enthalpy per unit mass. n_x, n_y, n_z represent the unit normal vector of the control body surface, and V_n represents the normal velocity of the control body surface.

To close the system of equations, it is necessary to introduce the ideal gas equation of state:

$$p = \rho RT \quad (4)$$

The relationship between the total energy per unit mass, total enthalpy per unit mass, and the density ρ , velocity, and pressure of the fluid is given as:

$$p = (\gamma - 1) \left[\rho E - \frac{1}{2} \rho (u^2 + v^2 + w^2) \right] \quad (5)$$

$$H = E + \frac{p}{\rho} \quad (6)$$

In addition,

$$\Theta_x = u\tau_{xx} + v\tau_{xy} + w\tau_{xz} + k \frac{\partial T}{\partial x} \quad (7)$$

$$\Theta_y = u\tau_{yx} + v\tau_{yy} + w\tau_{yz} + k \frac{\partial T}{\partial y} \quad (8)$$

$$\Theta_z = u\tau_{zx} + v\tau_{zy} + w\tau_{zz} + k \frac{\partial T}{\partial z} \quad (9)$$

where Θ_x , Θ_y , Θ_z express the heat conduction in the fluid, k is the thermal conductivity coefficient, and T is the temperature.

2.3. Calculation of wall heat flux

The wall heat flux is defined as [29] :

$$q_w = -k \nabla T \quad (10)$$

$$k = \frac{\mu}{(\gamma - 1) \text{Pr}} \quad (11)$$

where μ is molecular viscosity, γ denotes the specific heat ratio, and Pr is Prandtl number.

2.4. Numerical methods and boundary conditions

In this paper, RANS equation is used in turbulence simulation, and the RANS equation is solved by finite volume method. The second-order AUSMDV scheme is used for spatial discretization, the LU-SGS

Table 1

Error estimates.

Allowable error (%)	Grid resolution	CFL	Number of time steps	Allowable number of time steps	Reliability $R_s = \frac{n_{max}}{n}$
5	800×200	1	12×10^4	4×10^{10}	3.33×10^5
5	800×200	2	6×10^4	4×10^{10}	6.66×10^5

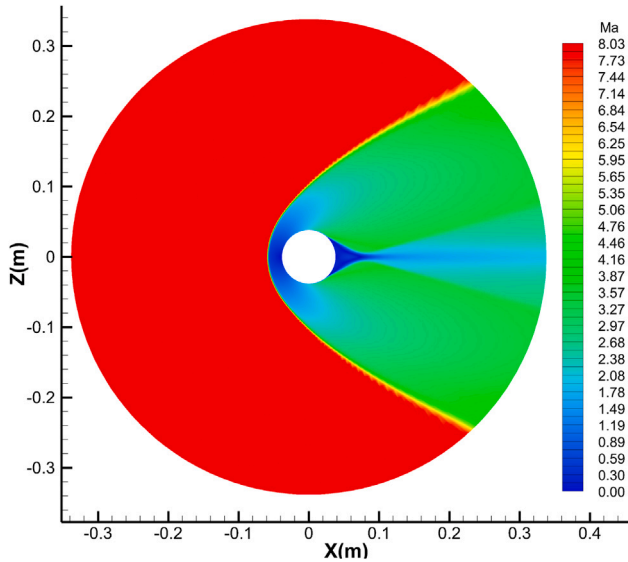
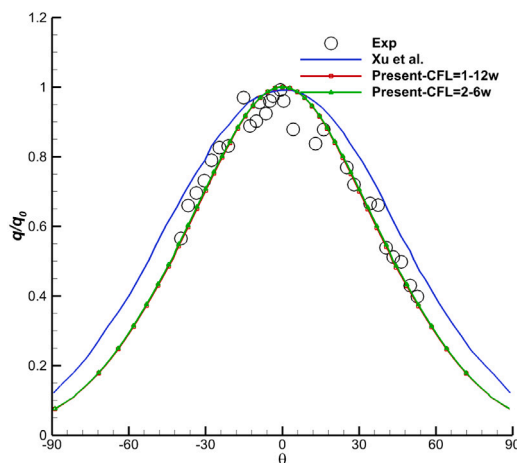


Fig. 2. The contours of Mach numbers around the cylinders.

method is used for time discretization, and the two-equation Menter's SST $k-\omega$ model [27,37] is used to describe the turbulence. The non-slip boundary and isothermal wall conditions are adopted for the wall, and the non-reflective boundary is adopted for the far-field. The detailed flow conditions are freestream unit Reynolds number $Re_\infty = 1.1 \times 10^7/m$, freestream Mach number $Ma_\infty = 6$, freestream pressure $P_\infty = 5529.1$ Pa, freestream temperature $T_\infty = 216.65$ K, wall temperature $T_w = 300$ K.



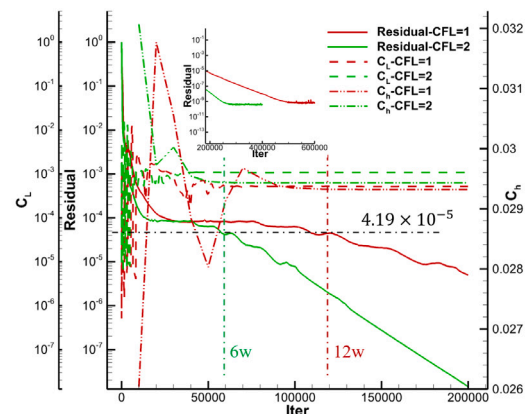
(a) The heat flux along the cylindrical surface

2.5. Validation and verification

2.5.1. Test 1: Hypersonic flow around a circular cylinder

A numerical example of hypersonic flow passing through a circular cylinder is selected as a benchmark problem to verify CFD solvers, in which the results of the present CFD code are compared with experimental data conducted by Wieting [38] and the numerical solution of Xu et al. [39]. The inflow condition for the air is freestream unit Reynolds number $Re_\infty = 4.816 \times 10^6/m$, freestream Mach number $Ma_\infty = 8.03$, freestream temperature $T_\infty = 124.94$ K, and wall temperature $T_w = 294.44$ K.

The computed contours of Mach numbers around the cylinders are shown in Figs. 2, and 3(a) shows that heat flux on the front surface of the cylinder, in which the present CFD codes present better coincidence of numerical results with experimental data than the numerical solution of Xu et al. [39]. As well known, at the end of the computational domain of supersonic flow, the results obtained are reliable or absolutely unreliable by using the same code but with different integration steps. Thus, using present code to calculate the reliability of the results in the supersonic steady flow is necessary under different Courant–Friedrichs–Levy number (CFL): $Courant = u \frac{\Delta t}{\Delta x}$, where u is the fluid velocity, Δt is the time step, and Δx is the mesh size in the direction of integration. The CFL number refers to the relative relationship between the time step and the space step. The larger is the CFL number, the longer is the integration step size, and the faster is the convergence speed. In short, due to the influence of CFL number on the numbers of integration steps, the cumulative errors are different. Thus before comparing the reliability of the results under different CFL numbers, we must judge their convergence separately. As seen in Fig. 3(b), the residual value does not change after 600 000 and 300 000 steps respectively, and it can be judged that the calculation has converged. Calculating the machine zero points is a simple but time-consuming method to judge the iterative convergence, which wastes more than half of the machine time to calculate the machine zero points after convergence. At the same time, it is obvious that the difference between the convergence of lift coefficient and that of heat flux is distinct. When the lift coefficient has fully converged, the heat flux under different CFL conditions is still



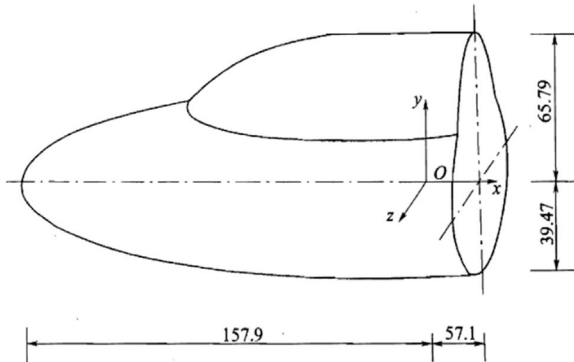
(b) Convergence curves of different monitoring values

Fig. 3. Calculation results of circular cylinder.

Table 2

Grid system information.

Grid name	Coarse grid	Medium grid	Fine grid
Total grid numbers	49.4 million	63.3 million	73.4 million

**Fig. 4.** Geometry of the double ellipsoids.

slowly decreasing. Therefore, it is a more direct and effective method to observe the convergence of aerodynamic heat data. The cylinder heat flux under different CFL numbers converges at 120 000 steps (CFL = 1) and 60 000 steps (CFL = 2) respectively. And then we use the same method of precision estimates described by Smirnov et al. [40,41] to compare the reliability of the results of 120 000 steps at CFL = 1 and that of 60 000 steps at CFL = 2 in Table 1. As it is seen from the table, the reliability (R_p) of the results which indirectly characterizes the accumulated error increases on increasing CFL number. The higher is the value, the lower is the error. [40,41].

2.5.2. Test 2: Double ellipsoids

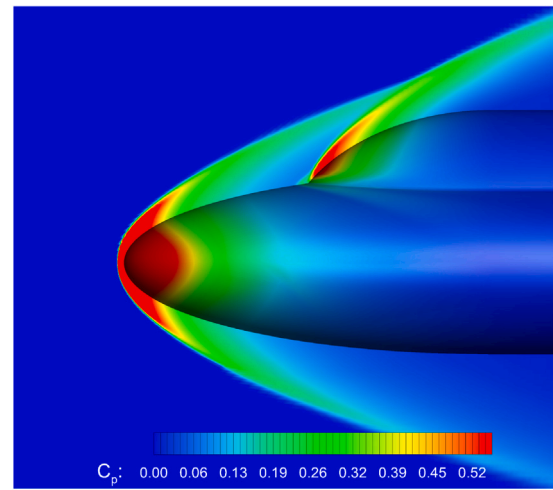
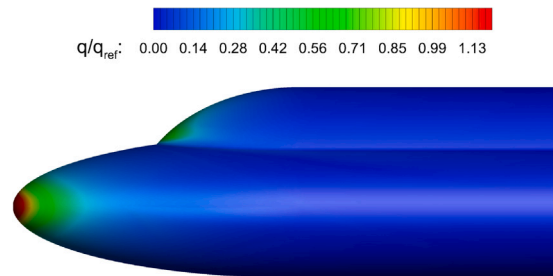
In order to valid the numerical approach and CFD code, the heat flux on the centerline of double ellipsoids is applied to discuss the code validation. The inflow conditions of the double ellipsoids shown in Fig. 4 are given in Ref. [42]. The wall heat flux distribution of the double ellipsoids is verified under the condition of $Ma_\infty = 8.04$ and freestream unit Reynolds number $Re_\infty = 1.13 \times 10^7/\text{m}$. The heat flux of the stagnation point is the reference value q_{ref} , which is used in determining the dimensionless heat flux data of surfaces with the same flow field. The dimensionless heat flux is also called the heat flux ratio, and has the form of q/q_{ref} ; hereinafter, this is referred to as the heat flux, and $q_{ref} = 568.4 \text{ (kw/m}^2\text{)}$.

An unstructured grid is used to calculate the double ellipsoid model in this paper, with a structural grid used near the wall. The contour of pressure coefficient is shown in Fig. 5, and Fig. 6 shows heat flux of the double ellipsoids, where it can be seen that heat flux is higher at the nose and the junction of the double ellipsoids, and is also enhanced at the shock wave interference position. Fig. 7 shows comparisons of heat flux along the centerline of the double ellipsoids given by the numerical simulations and from experimental data. As can be seen in Figs. 7(a) and 7(b), the numerical results are in good agreement with the experimental data.

The two tests above reveal that the CFD code in the present study can simulate the wall heat flux accurately and can be applied to predict the aerothermodynamic performance of the TSTO system.

2.6. Grid independence

In order to prove the grid size of the present computations is sufficient, the grid independency is performed by comparing the results of three grid scales. The detailed information of the grids is tabulated in Table 2.

**Fig. 5.** Contours of pressure coefficient C_p .**Fig. 6.** Contours of heat flux.

The windward centerline heat flux distributions on the WSRV of the three grid scales are shown in Fig. 8, where the reference heat flux q_{ref} equals to $1.6 \times 10^7 \text{ (w/m}^2\text{)}$. The results demonstrate that the medium grid has been able to ensure the independence of the grid, and the difference between the prediction results of the medium grid and the fine grid is small. Thus the medium grid is employed.

3. Results and discussion

Taking the TSTO model as an example, this paper focuses on the aerothermal phenomena of major critical parts at three reflected shock positions. The shock structure schematic of the WSRV and rocket used in the numerical simulations is shown in Fig. 9. An oblique shock induced by the nose of WSRV fuselage interacts with the leading edges of the canted fins and wings, and intersects with an incident shock wave attached to the head apex of the rocket and expansion waves. The incident shock generated by the rocket impinges on the WSRV, reflects on the two stages, and gradually weakens. Another part of the incident shock wave, which does not reflect on the two stages, interacts with the windward surface of the canted fins and wings. These interactions produce a shock wave/boundary-layer interaction (SWBLI), leading to high heat flux on the windward surface of the WSRV and upper surface of the rocket.

This paper presents a qualitative analysis and quantitative evaluation of the heating along the windward centerline and the axial stations around the various reflected shock waves of the two stages. Accordingly, the types of interstage interference near each reflected shock wave position, which cause various distributions of heat flux, are analyzed and divided into two categories.

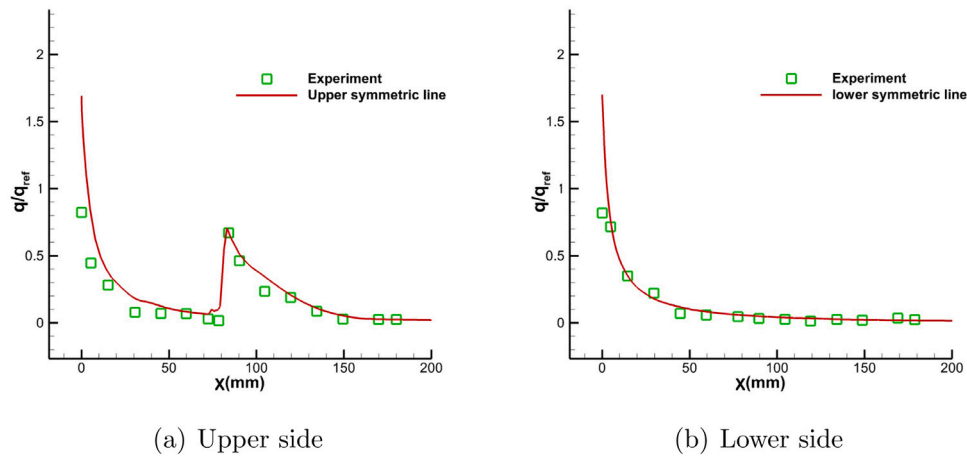


Fig. 7. Heat flux distributions along the centerline of the double ellipsoids.

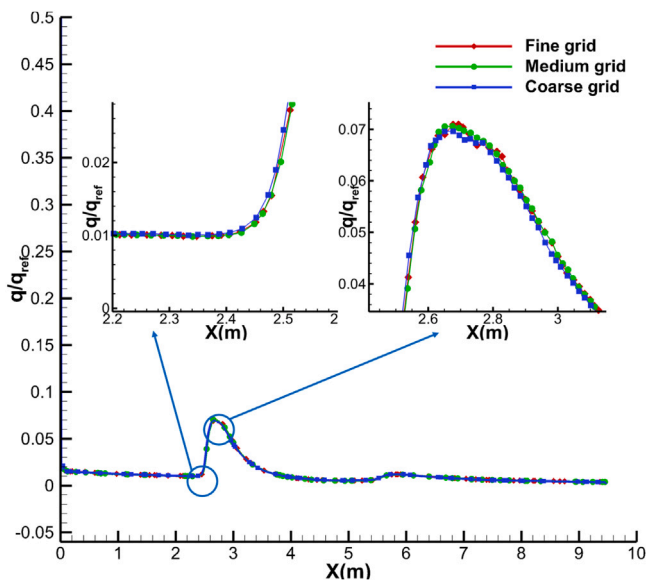


Fig. 8. Windward centerline heat flux distributions on the WSRV of different grids.

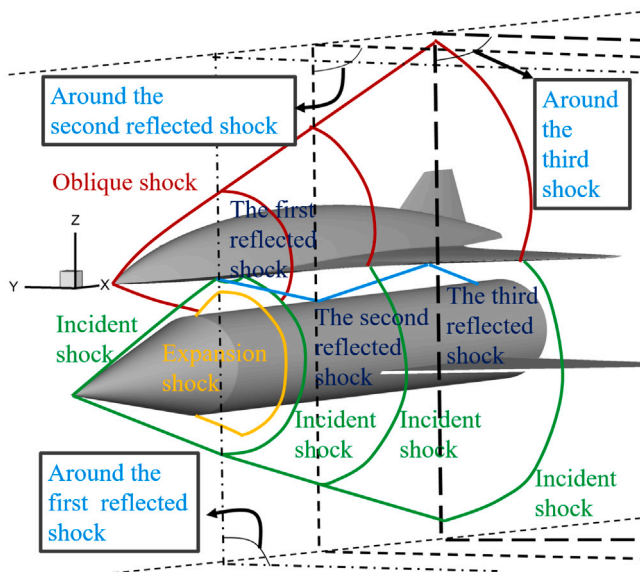


Fig. 9. Shock structure schematic.

3.1. Flow field characteristics of the global system and symmetric plane

Fig. 10 shows the global interstage environment for the TSTO system. The heat flux contours over the entire model are presented in Fig. 10(a). The complex wave structure generated by the shock/shock interaction is illustrated by the symmetry plane of density gradient $\nabla\rho$ and the contours of pressure coefficient C_p on the two stages in Fig. 10(b), and the contours of the density gradient $\nabla\rho$ at the symmetry plane are shown in Fig. 10(c). It is clear that high heat fluxes occur at the stagnation point of the WSRV, head of the rocket, leading edges, windward surfaces of the canted fins and wings, and rudder, as indicated in Fig. 10(a). These results show that the heat flux is higher at the locations of shock interference between the two stages. As shown in Fig. 10(b), some impinging shock waves, expansion waves, and reflected shock waves are generated, forming a complex flow pattern in the interaction region of the two stages. Note that an oblique shock that emanates from the nose of the WSRV fuselage interacts with the leading edges and the canted fins and wings. Between the WSRV and the rocket, an oblique shock induced by the nose of the WSRV windward surface intersects with an incident shock wave attached to the head apex of the rocket and the expansion waves generated at the rocket head-cylinder junction. The incident shock wave generated by the rocket then impinges on the WSRV, and is reflected onto the two stages, whereby the intensity of the reflected shock wave gradually weakens, as indicated in Fig. 10(c). The results illustrate that the shock wave reflects on the two stages at $X = 2.40$ m, 3.82 m, 5.45 m and 7.00 m, with peak values of heat flux at $X = 2.69$ m, 4.09 m, 5.80 m, and 7.50 m, respectively.

3.2. Aerothermal analysis of two stages around the first reflected shock wave

In Figs. 11(a) and 11(b), the contours of heat flux on the rocket upper surface and WSRV windward surface are magnified in the region of the two stages around the first reflected shock wave. There are relevant differences between the two enlarged views of heat flux distributions. Apparently, the range of high heat flux around the head of the rocket is greater than that around the WSRV. Additionally, the aerothermal load on the WSRV windward surface in the vicinity of the first reflected shock wave extends to the leading edge of the wing, forming a large “V” pattern that is discussed later, whereas that on the rocket upper surface is not obvious.

The heat flux along the windward centerline and at $X = 2.69$ m are shown in Figs. 12(a) and 12(b), and the skin friction coefficient profiles along the windward centerline and at $X = 2.69$ m are shown in Figs. 12(c) and 12(d). As displayed in Fig. 12(a), there is a distinct

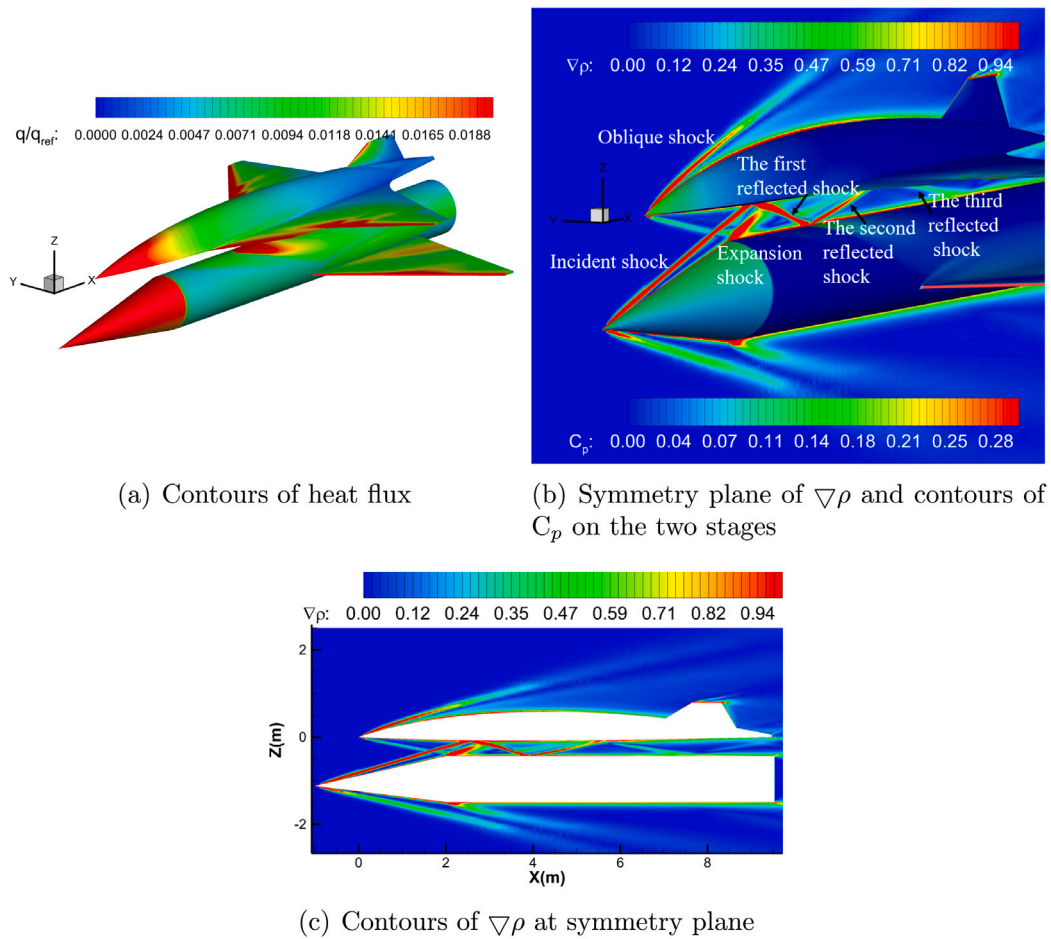


Fig. 10. Global interstage environment for the TSTO system.

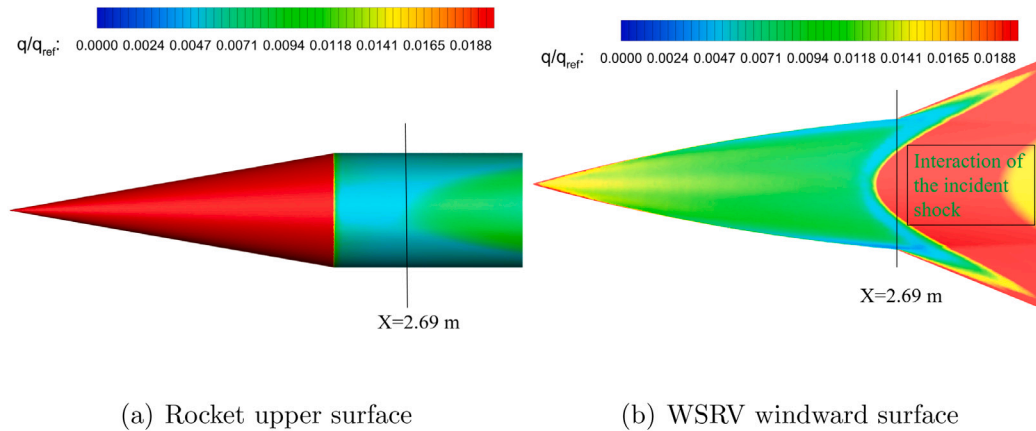


Fig. 11. Enlarged views of the distribution of heat flux around the first reflected shock.

trend in the heat flux on the windward surface of the WSRV with respect to the upper surface of the rocket. For the WSRV, from $X = 1.8$ – 4 m, the shock generated by the rocket impinges on the windward surface of the WSRV at $X = 2.40$ m, simultaneously introducing the first reflected shock wave and heating the surface by approximately 0.01, followed by a sudden jump. It is evident that $X = 2.69$ m corresponds to the peak value on the WSRV. The abrupt jump is observed to a level with a maximum of about 0.0648, which is approximately six times higher than that of the first shock reflection position at $X = 2.40$ m, followed by a slow drop to 0.011 at $X = 3.8$ m. For the rocket, from $X = 1.8$ – 4 m, the rocket head–cylinder junction at $X = 2$ m

immediately drops to a value of 0.0055 due to the expansion waves, and a small fluctuation occurs at $X = 2.69$ m. As demonstrated in Fig. 12(b), the aerothermal performance undergoes significant changes at different positions, and the value on the WSRV windward surface is much higher than on the rocket upper surface at all positions. For the WSRV windward surface, higher values appear closer to the centerline, and a value of 0.0648 is attained at $Y = 0$ m. The rocket upper surface exhibits similar behavior as the WSRV windward surface, with a slight increase to roughly 0.009. As illustrated in Figs. 12(c) and 12(d), the skin friction coefficient $C_f = 0.0063$ of the WSRV windward surface at $Y = 0$ m is much higher than that of other positions from the perspective

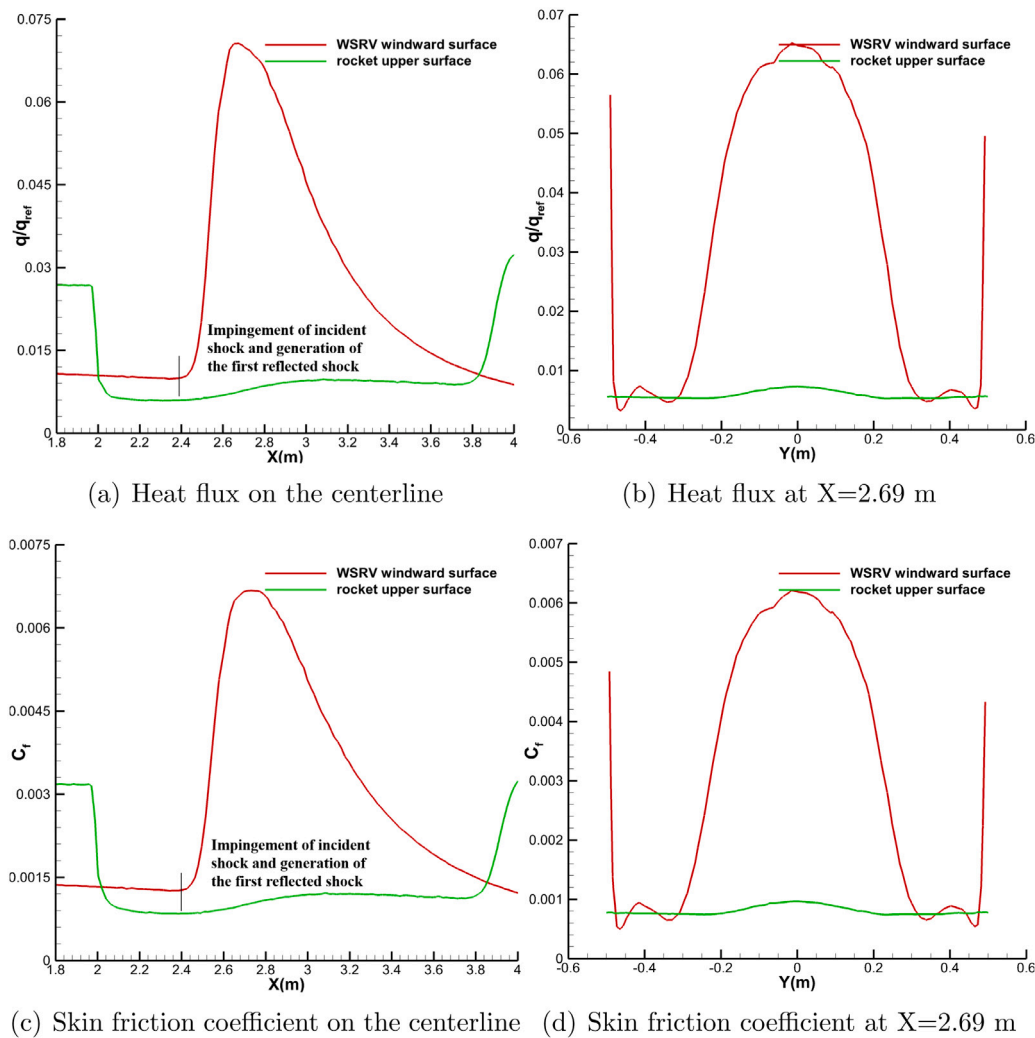


Fig. 12. Heat flux/skin friction coefficient profiles around the first reflected shock.

of flow direction and cross-flow direction, and C_f is positive at all positions, indicating that no separation bubbles are generated in these regions. The incident shock wave generated by the head apex of the rocket is of moderate intensity. The viscosity and the shear stress in the boundary layer play important roles in the flow at this Reynolds number. Therefore, the shear stress in the boundary layer can resist the adverse pressure gradient introduced by the incident shock wave. Besides the shear stress, the velocity distribution in the boundary layer is also influential. When the fluid passes over the windward surface of the WSRV with convex curvature, as seen in Fig. 13, the velocity gradient and the pressure gradient increase because of the barotropic gradient and expansion motion [43,44]. In addition, momentum is transported from the external high-speed region to the internal low-speed region, while the "full velocity distribution" in the boundary layer carries more momentum and is relatively unaffected. For the above reasons, the resistance of the boundary layer to the adverse pressure gradient increases. There is no separation region in the boundary layer, which accompanies higher shear stress and C_f . Moreover, the velocity of the boundary layer decreases after the shock wave, causing a similar increase in pressure, and the heat flux corresponds to the trend in the skin friction coefficient. In the SWBLI produced by impingement of the incident shock wave attached to the head apex of the rocket on the WSRV windward surface, the incident shock wave is deflected in the boundary layer, and its intensity gradually decreases in the process of passing from the boundary layer to the sound velocity line. The pressure rise after the incident shock wave propagates upstream through the

very thin subsonic region in the boundary layer, thus affecting the thickness of the subsonic layer in the boundary layer, and resulting in a set of external compression wave systems that converge into the first reflected shock wave, rather than the separated shock wave formed by a strong incident shock.

As presented in Fig. 14, in combination with Figs. 10(b) and 10(c), it is clear that there is an incident shock wave attached to the head apex of the rocket. This is very close to the rocket head surface, and causes elevated heat flux on the head of the rocket. Expansion waves at the rocket head-cylinder junction decrease the pressure and lead to lower heat flux. The shock induced by the nose of the WSRV windward surface is observed to impinge on the upper surface of the rocket, which brings about SWBLI, leading to a slight increase in the heat flux from $X = 2.40$ – 2.69 m, as demonstrated in Figs. 11(a), 12(a) 12(b). For the WSRV, an oblique shock induced by the nose of its windward surface, very close to the WSRV head, results in a rise in the heat flux in front of the head. The elevated heating of the WSRV windward surface in the flow and axial directions, shown in Figs. 12(a) and 12(b), results from the SWBLI caused by the incident shock, which accompanies the first reflected shock wave that interferes with the expansion waves. The intersection line between the interaction of the incident shock wave that is not reflected on the two stages and the WSRV windward surface forms a large "V" pattern that extends to the leading edge of the wings; an oblique shock by itself does not work within these regions of the WSRV, as demonstrated in Fig. 11(b).

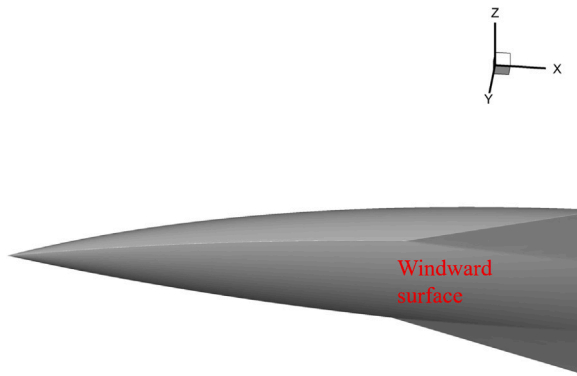


Fig. 13. Enlarged view of WSRV windward surface.

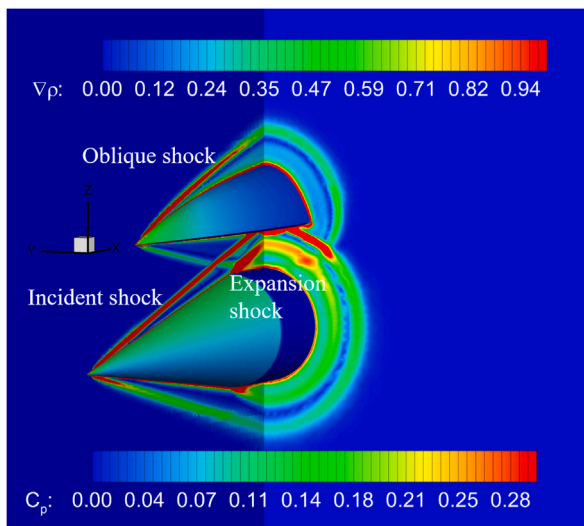


Fig. 14. Enlarged view of three-dimensional slices at $X=2.69$ m around first reflected shock.

Although the shock intensity considered in this paper is moderate and there is no separation, if the rocket head forms a strong shock wave, separation may occur depending on the design of the WSRV head.

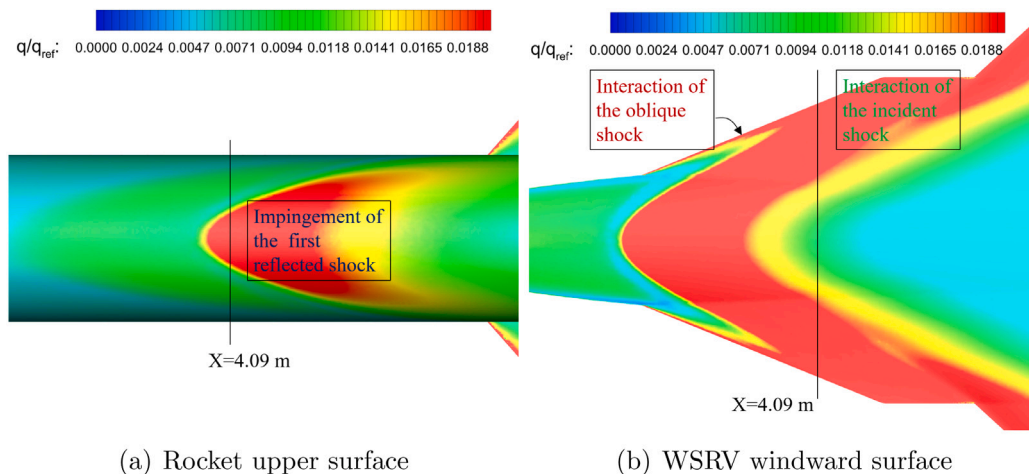


Fig. 15. Enlarged views of the distribution of heat flux around the second reflected shock.

3.3. Aerothermal analysis of two stages around the second reflected shock wave

Figs. 15(a) and 15(b) display enlarged views of the contours of heat flux on the rocket upper surface and the canted fin windward surface around the second reflected shock wave. Obviously, the heat flux in the rocket around the second reflected shock wave is dramatic. In addition, the heating load in the vicinity of the middle of the windward surface of the plane gradually returns to a low level. The values on the canted fin windward surface become higher towards the leading edges.

To enable a detailed quantitative prediction, the heating along the windward centerline of the two stages around the second reflected shock wave and at the axial station along the model is now examined. The heat flux and skin friction coefficient profiles along the windward centerline are shown in Figs. 16(a) and 16(b), and the heating and skin friction coefficient profiles for axial stations of $X = 4.09$ m are presented in Figs. 16(c) and 16(d).

Fig. 16(a) illustrates the discrepancy in the heat flux between the windward side of the WSRV and the upper surface of the rocket around the second reflected shock. For the WSRV from $X = 3.8$ – 5.4 m, the value drops to 0.005 at $X = 5.4$ m. For the rocket from $X = 3.8$ – 5.4 m, the first reflected shock impinges on the upper surface of the rocket at $X = 3.82$ m and the heating reaches approximately 0.0056, inducing a second reflected shock wave. An increase can be observed over the remainder of the rocket, with a peak value of about 0.03164 at $X = 4.09$ m, followed by a sudden drop to 0.01 at $X = 5.4$ m. As illustrated in Fig. 16(c), the rocket upper surface exhibits the opposite behavior to the WSRV windward surface in this region. The aerothermal value of the WSRV windward surface is much higher than that of the rocket upper surface, except for a region in the vicinity of the second reflected shock wave. For the canted fin windward surface, higher values can be found closer to the leading edges of the canted fins, with a value of 0.07 at both $Y = -1$ m and 1 m. This decreases considerably to roughly 0.01 at $Y = 0$ m. For the rocket upper surface, there is an increase in heat flux around the second reflected shock wave, with a maximum of about 0.03164. As demonstrated in Figs. 16(b) and 16(d), the skin friction coefficient of the rocket upper surface, $C_f = 0.0035$, is much higher than that of the flow and lateral positions, and C_f is greater than zero at all positions, signifying that no separation zones appear in these regions. Thus, the intensity of the first shock wave and the incident shock wave is not sufficient to produce separation on the rocket upper surface and canted fin windward surface, respectively.

As displayed in Fig. 17, coupled with Figs. 10(b) and 10(c), the increase in heating is the result of the SWBLI induced by the interaction of the first reflected shock wave with the rocket upper surface, which induces the second reflected shock wave. At the leading edges of the

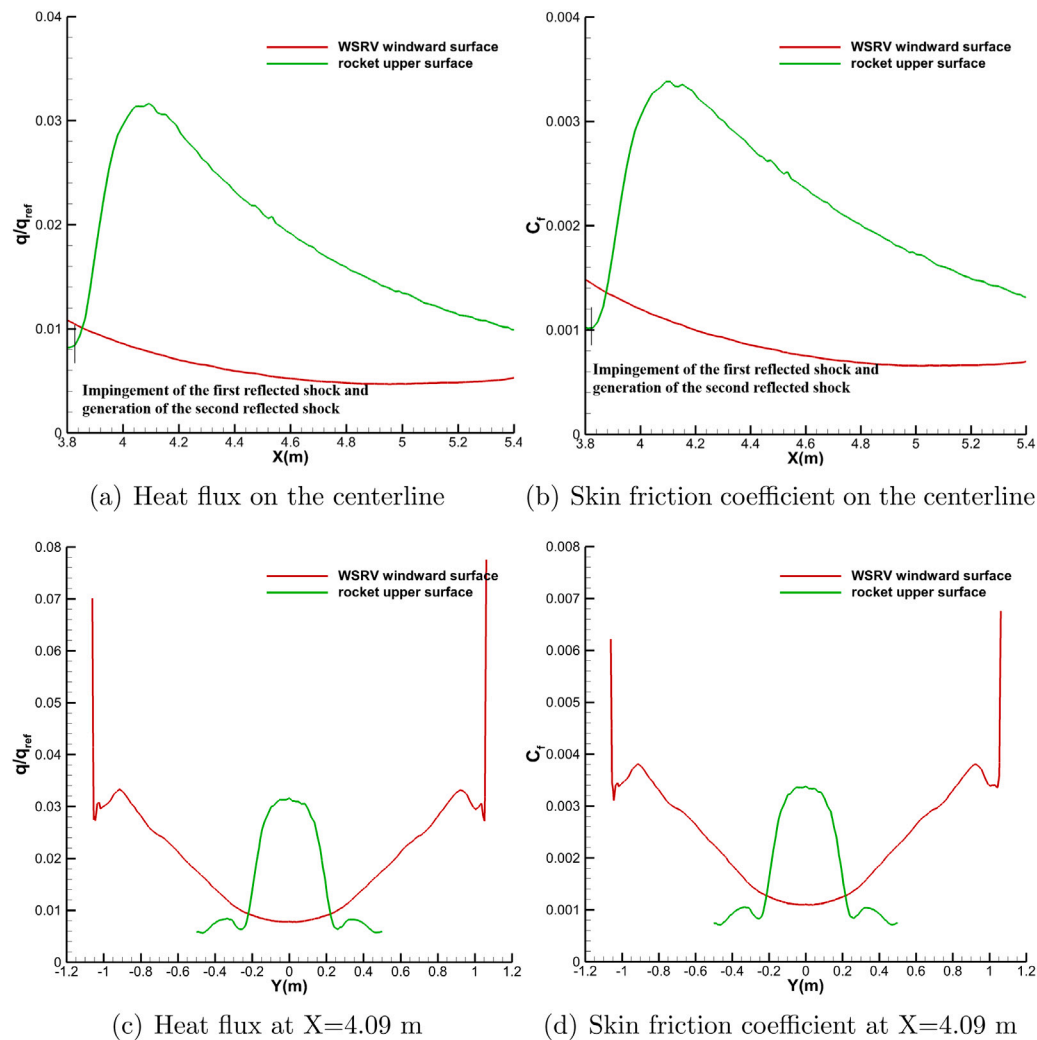


Fig. 16. Heat flux/skin friction coefficient profiles around the second reflected shock.

canted fin, the elevated heating is caused by the interaction of the oblique shock generated by the nose of the WSRV fuselage with the leading edges of the canted fin. For the canted fin windward surface close to the leading edges, elevated thermal values arise from the SWBLI induced by the impingement of the incident shock wave that does not reflect on the two stages. This is probably due to an improper interstage distance, the sweep angle of the canted fin, and its width. Therefore, it is suggested that the above design factors should be considered to prevent the shock wave generated by the rocket from interfering with the aircraft when designing the TSTO model.

3.4. Aerothermal analysis of two stages around the remainder of the model

Figs. 18(a) and 18(b) show zoomed views of the heat flux over the remainder of the rocket upper surface and windward surface of the wing. High heat fluxes appear along the windward centerline of the WSRV, windward surface of the wing, and at the leading edges of the wings of the WSRV and the rocket. The heat flux in areas close to the leading edges of the wing is higher than along the windward centerline of the WSRV.

To quantitatively describe the distribution trend, the heating along the windward centerline of the two stages around the remainder of the model and at the axial station is now investigated. The heat flux and skin friction coefficient profiles along the windward centerline are shown in Figs. 19(a) and 19(b), and the heating profiles at X = 5.80 m and 7.50 m are presented in Figs. 19(c) and 19(d). Skin friction

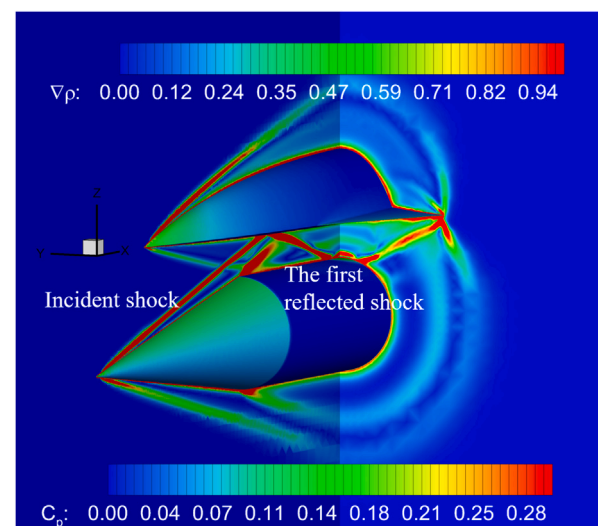


Fig. 17. Enlarged view of three-dimensional slices at X = 4.09 m around the second reflected shock.

coefficient profiles at X = 5.80 m and 7.50 m are shown in Figs. 19(e) and 19(f).

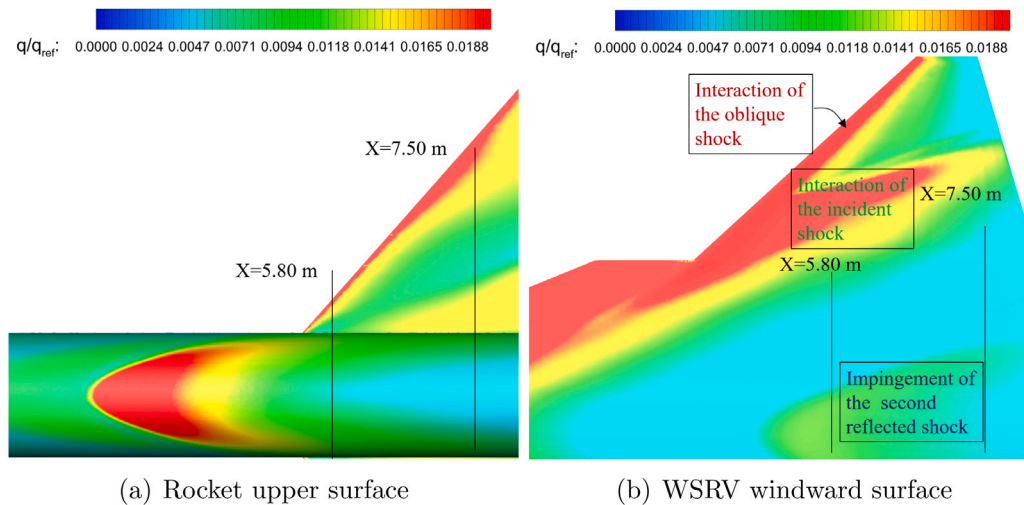


Fig. 18. Enlarged views of the distribution of heat flux around the third reflected shock.

Fig. 19(a) indicates that the reflected shock wave becomes weaker and the level of heating fluctuates slightly, although it remains approximately constant. In other words, the different magnitudes of opposite fluctuations have diminished significantly. The trend in the skin friction coefficient is similar to that of the heat flux along the windward centerline, as shown in Fig. 19(b). As illustrated in Fig. 19(c), the behavior of the rocket upper surface corresponds to that of the windward surface of the wing (except for the windward centerline of the WSRV). Different trends exist from $X = -0.5$ – 0.5 m. For the windward surface of the wing, higher values can be observed closer to the leading edges of the wing, and a value of 0.13 is attained at both $Y = -2$ m and 2 m. This fluctuates before falling to roughly 0.01 at $Y = -0.5$ m and 0.5 m, followed by a slight increase to 0.02 at the centerline. At the rocket upper surface, the heat flux undergoes fluctuations while decreasing from 0.15 to 0.015, and then changes slightly at $Y = -0.5$ m and 0.5 m. In Fig. 19(e), the variation in the axial friction coefficient profile is the same as that in the heat flux. The skin friction coefficient C_f of the WSRV wing windward surface decreases from 0.011 to 0.0008 and then rises to 0.0013, while that of the rocket upper surface decreases from 0.009 to 0.001. Figs. 19(d) and 19(f) indicate that the relative trends are similar to those in the previous position at $X = 5.80$ m, respectively, although their values are much lower. In this region, the reflected shock wave is already very weak, and is thus even less prone to separation.

As indicated in Figs. 20(a) and 20(b), together with Figs. 10(b) and 10(c), the elevated heating on the WSRV windward surface is the result of complex interactions. The second reflected shock wave caused by the previous position at 3.82 m impinges and reflects on the windward centerline of the WSRV, forming a third reflected shock wave and bringing about SWBLI with a rise in heat flux. For the wing windward surface, the interaction of an impinging shock wave generated at the head apex of the rocket also induces SWBLI, leading to a peak value of 0.04. For the leading edges of the WSRV wing, the impingement of an oblique shock induced by the nose of the WSRV can result in a maximum value of 0.13, which is much higher than that of the leading edges of the canted fin at all positions. For the leading edges of the rocket wing, the elevated aerothermodynamic magnitude stems from interference with the shock wave attached to the head apex of the rocket.

In this area, the rocket and WSRV still suffer some interference, which is unfavorable to the whole TSTO system. Therefore, a reasonable design of the TSTO model should consider the interstage distance, sweep angle of the wing, and wingspan.

3.5. Influence of the angle of attack on interstage thermal environment

In this section, we study the effects of different angles of attack on the aerothermodynamics of the two stages. The angles of attack considered herein are -6° , 0° , and 2° .

As shown in Fig. 21, the distributions of heat flux on the windward surface of the WSRV at all three angles of attack are V-shaped, and those at 0° and 2° are very similar. However, as the angle of attack decreases to -6° , the range of the V-shaped distribution of heat flux on the windward surface of the WSRV gradually narrows, and the level of heat flux on its surface also declines. As shown in Fig. 22, the distributions of heat flux of the rocket at the different attack angles are the opposite to those of the WSRV, with the greatest heat flux on the upper surface of the rocket occurring at an angle of attack of -6° and the smallest occurring at 2° . These differences in the heat flux on the rocket at various attack angles are the result of changes to the shock wave generated by the rocket itself, rather than that of the interstage interaction. In addition, as displayed in Fig. 23, the reflection positions of shock waves at the different angles of attack are almost identical, and the intensity of the shock decreases with the reflection of shock waves.

As the distribution of heat flux at the -6° angle of attack is significantly different from that at 0° , the flow fields of these two angles of attack are analyzed in further detail. As illustrated in Fig. 24, the decrease in intensity of the incident shock of the rocket at -6° weakens the SWBLI resulting from the interaction of the incident shock of the rocket with the windward surface of the canted fin, as well as the shock/shock interaction of the incident shock with the oblique shock wave of the WSRV. Thus, the heat flux on the windward surface of the canted fin corresponds to the trend in the intensity of shock interactions mentioned above. Additionally, as the attack angle changes, the interference areas of the incident shock wave with the windward surface of the canted fin become smaller, narrowing the range of the V-shaped distribution of heat flux. At the same time, the first reflected shock wave generated by the interaction of the incident shock with the windward surface of the WSRV produces SWBLI on the upper surface of the rocket, thus increasing the heat flux on the rocket surface. As demonstrated in Fig. 25, the strength of the second reflected shock wave and the incident shock wave of the rocket at the -6° angle of attack are weaker than those at 0° , resulting in reduced heat flux on the windward surface of the WSRV and the wings.

These results are instructive for the design of TSTO systems, because the range over which the shock impinges on the walls at different angles of attack and the variation of shock intensity not only affect the aerothermodynamic performance of the TSTO, but also influence the staging separation problems.

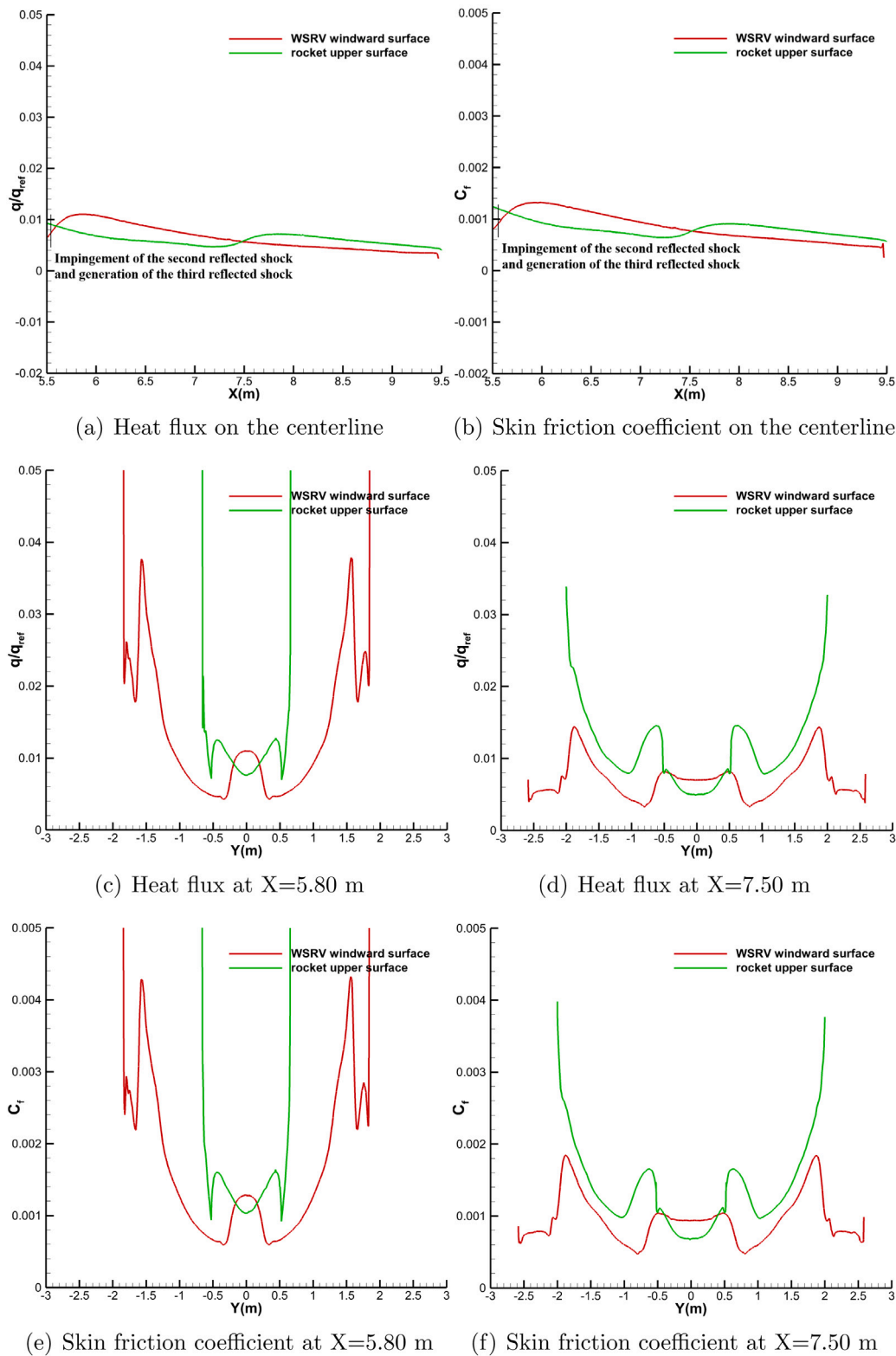


Fig. 19. Heat flux/skin friction coefficient profiles around the third reflected shock.

4. Summary and conclusions

The purpose of the study was to investigate the strong thermal performance of the complex structure of a wide-speed-range vehicle combined with a rocket with wings. Special attention has been paid to

the phenomena of the shock/shock and shock/boundary-layer interactions near the reflected shock locations between the wide-speed-range vehicle and the rocket, and the surface heat flux has been comprehensively analyzed. The conclusions to this study can be summarized as follows:

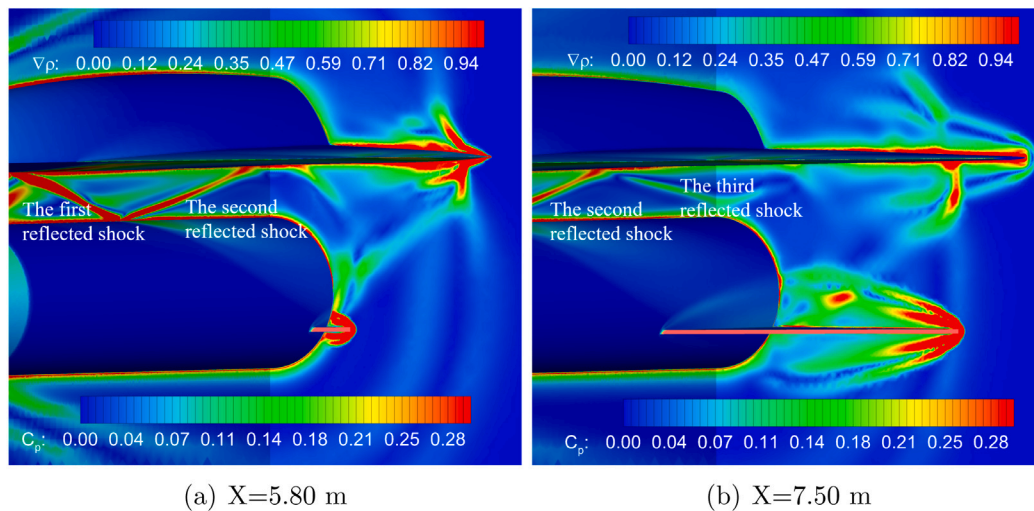


Fig. 20. Enlarged view of three-dimensional slices around the third reflected shock.

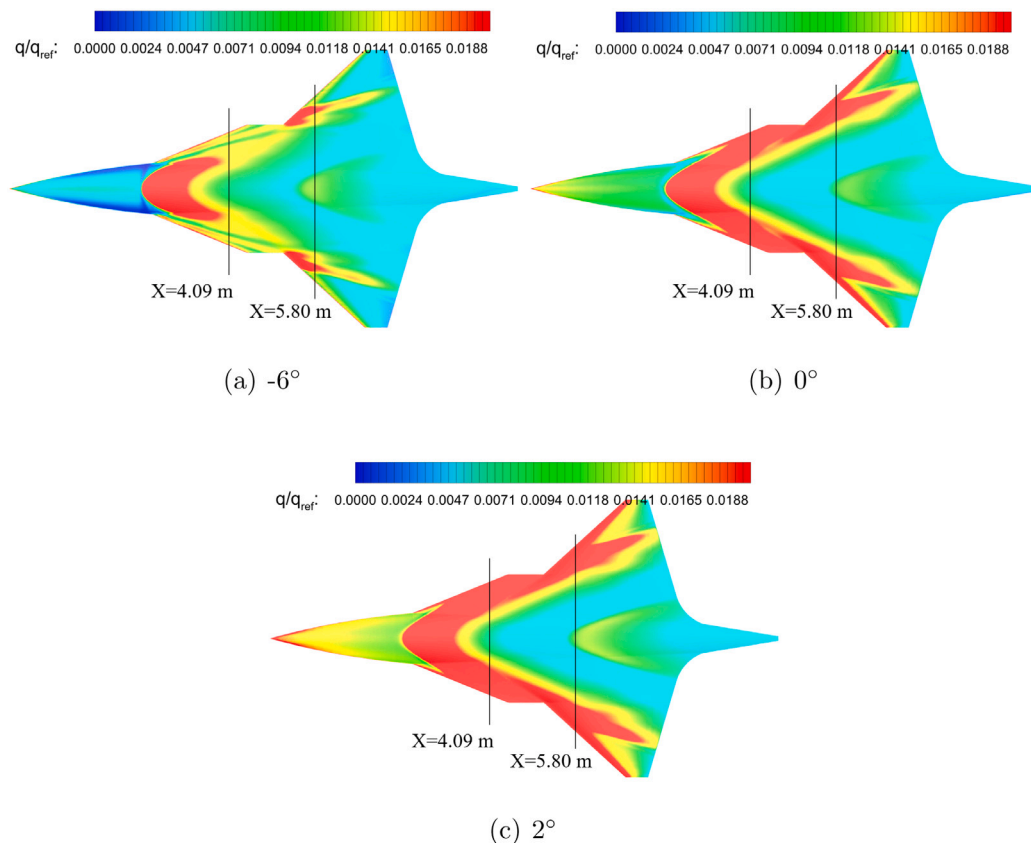


Fig. 21. Contours of heat flux on windward surface of WSRV at different angles of attack.

(1) The incident shock wave generated by the head apex of the rocket constantly reflects between the two stages, inducing shock wave/boundary-layer interaction and giving rise to elevated heat flux. The discrepancy in the heat flux between the windward side of the wide-speed-range vehicle and the upper surface of the rocket is very large at the front reflection positions, but there is little difference in the rear reflection positions as the reflected shock gradually weakens. The moderate incident shock and weak reflected shock are primarily responsible for the lack of boundary-layer separation, although the expansion caused by the convex curvature of the head of the wide-speed-range vehicle also contributes to this, with relatively high shear

stress and skin friction coefficient. Besides, the trend in the heat flux corresponds to that of the skin friction coefficient, because the increase in pressure in the boundary layer is blocked by the reflected shock waves.

(2) The part of the incident shock wave that is not reflected on the two stages interacts with the windward surface of the canted fins and wings, producing shock wave/boundary-layer interaction and a V-shaped heat flux distribution that extends to the leading edges of the wings.

(3) As the angle of attack decreases, the intensity of the incident shock and the reflected shock gradually becomes weak, leading to a

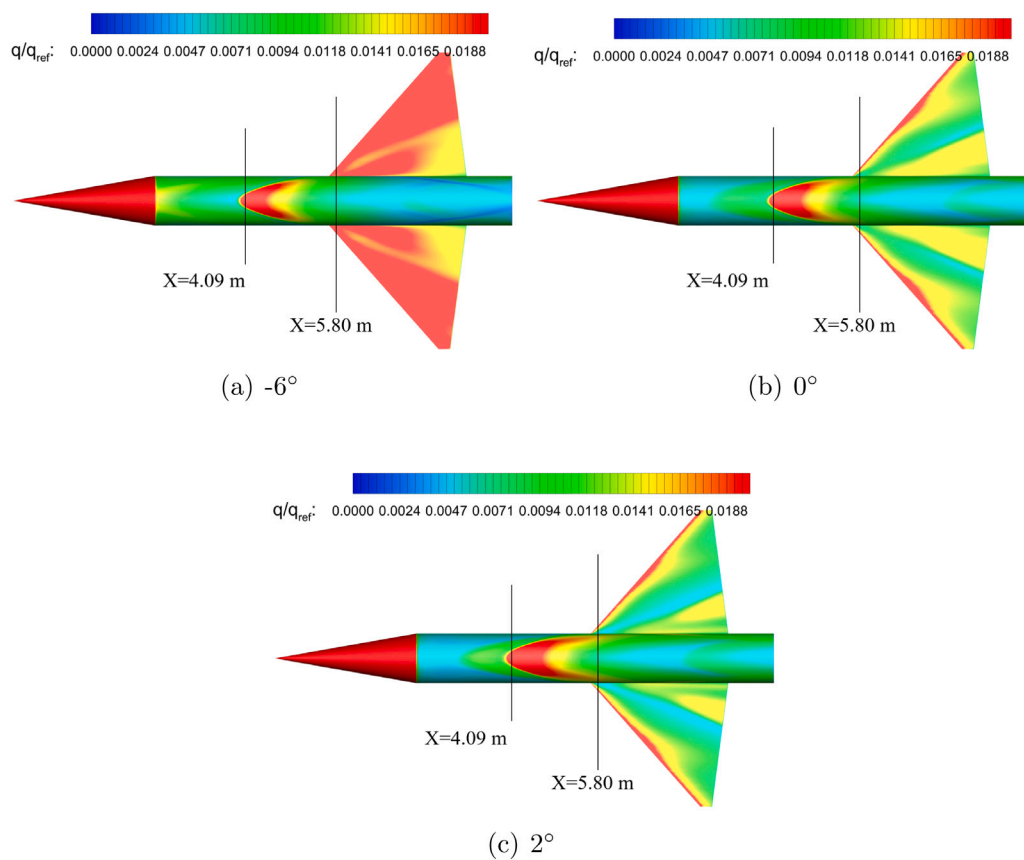


Fig. 22. Contours of heat flux on upper surface of rocket at different angles of attack.

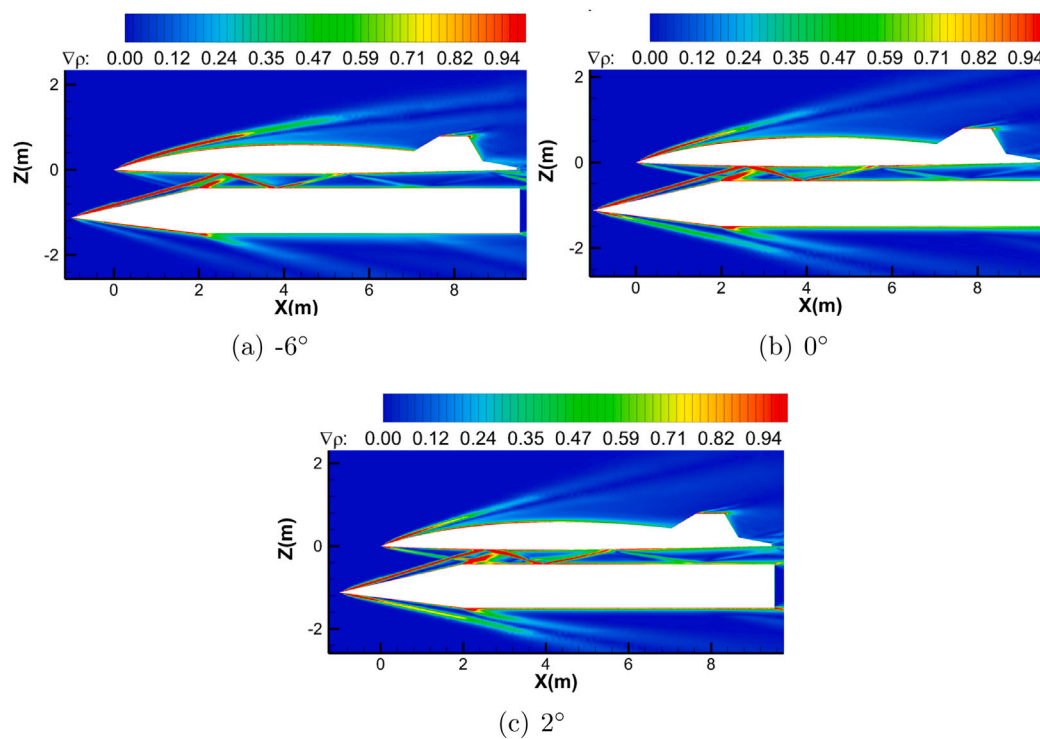


Fig. 23. Symmetry plane of $\nabla\rho$ at different angles of attack.

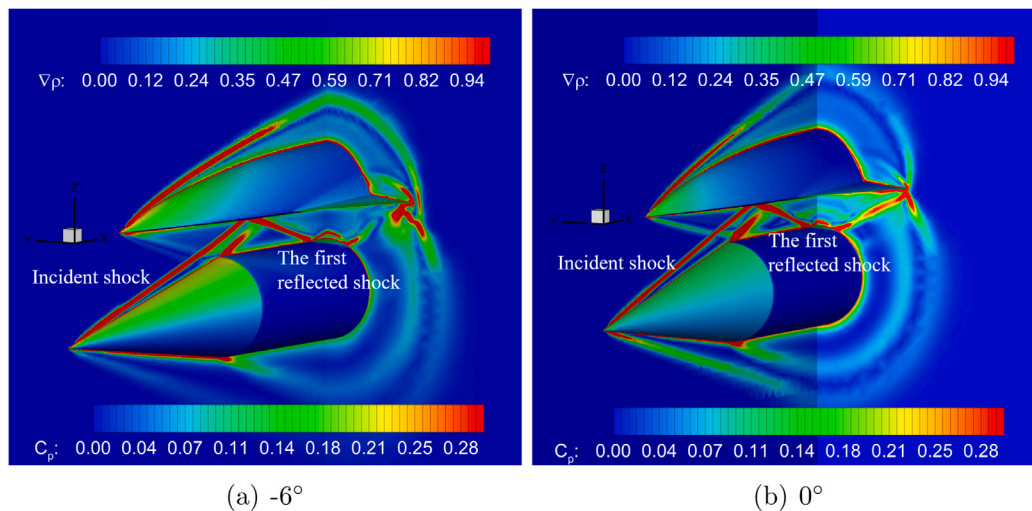


Fig. 24. Comparison of the enlarged view of three-dimensional slices at $X = 4.09$ m at -6° and 0° angles of attack.

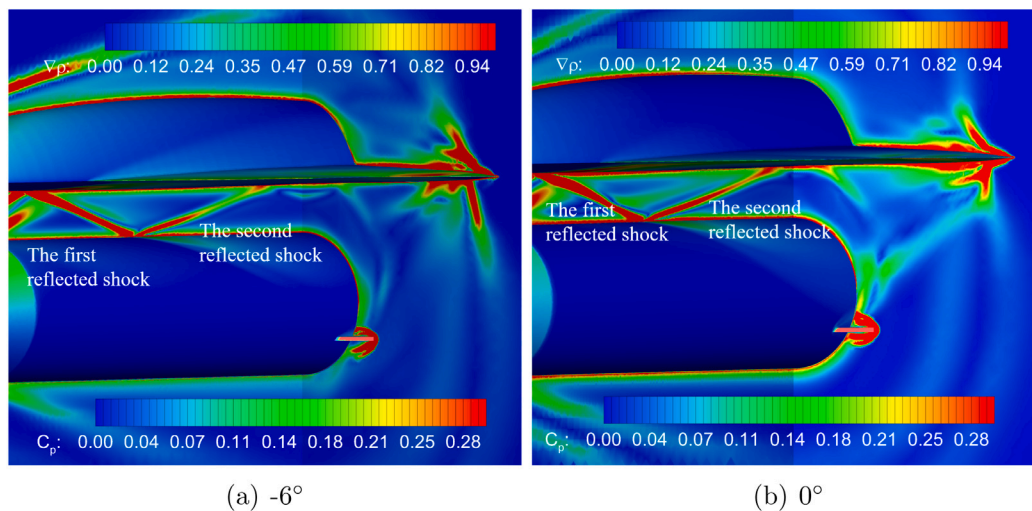


Fig. 25. Comparison of the enlarged view of three-dimensional slices at $X = 5.80$ m at -6° and 0° angles of attack.

reduction in the level of heat flux on the windward surface of the wide-speed-range vehicle. The range over which the incident shock impinges the walls becomes narrower, corresponding to the smaller distribution of heat flux.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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